A Few Notes on the Principle of Least Action and Newton's Second Law of Motion

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1 Introduction

The principle of least action, which is also known as the stationary-action principle, is a variational principle that, when applied to the action of a mechanical system, amazingly yields the equations of motion for that system. The principle states that the trajectories (i.e. the solutions of the equations of motion) are stationary points of the system's action functional [1]. One of the many amazing aspects of the principle is that it can be used to derive Newton's Second Law, Lagrangian and Hamiltonian equations of motion, and even general relativity (see Einstein–Hilbert action [2]).

These notes explore a bit of the connection between the Principle of Least Action [7] and Newton's Second Law [4].

2 Finding the Minimum of a Function

The story starts with how we use calculus to find the minimum of a function. In particular, suppose we have a function $f(x)$. We know that if we move a small distance from x, say ϵ, we know that $x \to \epsilon \Rightarrow f(x) \to f(x+\epsilon)$.

We also know that we can evaluate $f(x+\epsilon)$ using a Taylor series [8]. In particular

$$f(x+\epsilon) = f(x) + \epsilon\frac{df}{dx} + \frac{\epsilon^2}{2!}\frac{d^2f}{dx^2} + \frac{\epsilon^3}{3!}\frac{d^3f}{dx^3} + \cdots \qquad \text{\# definition of the Taylor series for } f(x+\epsilon)$$

Now if we define $\Delta f = f(x+\epsilon) - f(x)$ we see that

$$\begin{aligned}
\Delta f &= f(x+\epsilon) - f(x) && \text{\# definition of } \Delta f \\
&= \left[f(x) + \epsilon\frac{df}{dx} + \frac{\epsilon^2}{2!}\frac{d^2f}{dx^2} + \frac{\epsilon^3}{3!}\frac{d^3f}{dx^3} + \cdots\right] - f(x) && \text{\# substitute Taylor series for } f(x+\epsilon) \\
&= \epsilon\frac{df}{dx} + \frac{\epsilon^2}{2!}\frac{d^2f}{dx^2} + \text{higher order terms} && \text{\# } f(x) \text{ cancels} \\
&= 0 + \frac{\epsilon^2}{2!}\frac{d^2f}{dx^2} + \cdots && \text{\# } \frac{df}{dx} = 0 \text{ at } f\text{'s minimum} \\
&= \frac{\epsilon^2}{2!}\frac{d^2f}{dx^2} + \cdots && \text{\# simplify} \\
&\approx \frac{\epsilon^2}{2!}\frac{d^2f}{dx^2} && \text{\# } \epsilon \text{ is close to zero so we can ignore higher order terms}
\end{aligned}$$

So we can say that $\Delta f \sim \epsilon^2$, that is, the difference is a second order effect. Said from the perspective of first-order variation we can say that the first-order variation in the value of f is constant in some neighborhood around f's stationary point (that is, where the gradient of f equals zero).

So in summary we know that for normal functions f at the minimum

$$\Delta f = f(x+\epsilon) - f(x) = 0 \tag{1}$$

at least to the first order in ϵ. The constraint specified by Equation 1 will become important when we study $\delta\mathcal{S}$.

An important (or maybe, the most important) difference between the approach taken by the Principle of Least Action and Newton's Second Law is that the Principle of Least Action looks at the value of an entire *path*, where as Newton's Second Law looks at points along the path. As we will see, these two approaches turn out to be (amazingly) equivalent.

3 The Principle of Least Action

The action, denoted $\mathcal{S}$, is defined to be the sum of the difference between a particle's kinetic energy [3] and its potential energy [5] at every point along its path. The Principle of Least Action states that the path for which the sum of these differences has the minimum sum is the actual path the particle will take. To show that this is the case all of the proofs that I have seen rely on the fact that the Principle of Least Action and Newton's Second Law ($F = ma$) are equivalent. These notes show this result below.

To give a bit more formal definition of the Principle of Least Action:

Definition 3.1. The Principle of Least (Stationary) Action: The path of a particle is the one that yields a stationary value of the action. A stationary point of some function is a point where its gradient is equal to zero. In other words, the first-order variation in the value of the function is constant in some neighborhood around the stationary point. As we will see below, this condition can be expressed as $\delta\mathcal{S} = 0$.

Consider the setup shown in Figure 1 and imagine that the actual path is represented by the function $x(t)$. That is, $x(t)$ represents the path of a particle that starts out at position x_1 at time t_1 and ends up at x_2 at time t_2. Then what we want to do is to look at another path that is some small deviation away from $x(t)$ (we know from Equation 1 that this will be close to $x(t)$). We hope that if we look at the change in the action of the path with the small deviation that change should be close to zero as well.

As we can see from Figure 1 we call the small deviation $\eta(t)$ and we call the deviated path $x(t) + \eta(t)$.

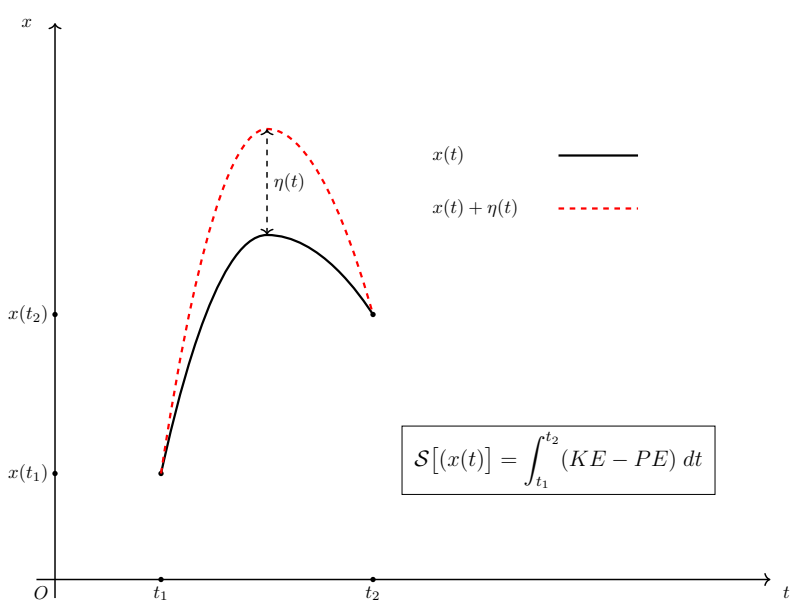

Figure 1: Principle of Least Action Setup

Now, consider a particle moving in one dimension in a uniform gravitational field g with potential energy $V[x(t)]$[1]. In this case $V[x(t)] = m \cdot g \cdot x(t)$. So

$$\begin{aligned}
\mathcal{S}\big[(x(t)\big] &= \int_{t_1}^{t_2} (KE - PE)\, dt && \text{\# definition of } \mathcal{S}\big[(x(t)\big] \\
&= \int_{t_1}^{t_2} \left[\frac{1}{2}m\left(\frac{d}{dt}x(t)\right)^2 - V\big[x(t)\big]\right] dt && \text{\# } KE = \tfrac{1}{2}m\left(\frac{d}{dt}x(t)\right)^2,\ PE = V\big[x(t)\big]
\end{aligned}$$

Hopefully simplifying by setting $x = x(t)$ we can see that

$$\mathcal{S}(x) = \int_{t_1}^{t_2} \left[\frac{1}{2}m\left(\frac{dx}{dt}\right)^2 - V(x)\right] dt \tag{2}$$

Now define $\delta\mathcal{S} = \mathcal{S}(x+\eta) - \mathcal{S}(x)$. We know that the path taken by the system, $x(t)$, has a stationary action ($\delta\mathcal{S} = 0$) under small changes to the configuration of the system. Said another way, we know that $\delta\mathcal{S} = 0$ to the first order in η (here again x is shorthand for $x(t)$ and η is shorthand for $\eta(t)$). So now we can expand $\mathcal{S}(x+\eta)$ as follows:

$$\begin{aligned}
\mathcal{S}(x+\eta) &= \int_{t_1}^{t_2} \left[\frac{1}{2}m\left(\frac{d}{dt}(x+\eta)\right)^2 - V(x+\eta)\right] dt && \text{\# substitute } x+\eta \text{ for } x \\
&= \int_{t_1}^{t_2} \left[\frac{1}{2}m\left(\frac{dx}{dt} + \frac{d\eta}{dt}\right)^2 - V(x+\eta)\right] dt && \text{\# derivative is a linear operator}
\end{aligned} \tag{3}$$

Next notice that

$$\begin{aligned}
\left(\frac{dx}{dt} + \frac{d\eta}{dt}\right)^2 &= \left(\frac{dx}{dt}\right)^2 + \left(\frac{d\eta}{dt}\right)^2 + 2\left(\frac{dx}{dt}\right)\left(\frac{d\eta}{dt}\right) && \text{\# multiply out} \\
&= \left(\frac{dx}{dt}\right)^2 + 2\left(\frac{dx}{dt}\right)\left(\frac{d\eta}{dt}\right) && \text{\# } \left(\frac{d\eta}{dt}\right)^2 \approx 0 \text{ (Section 2)}
\end{aligned}$$

Next want to expand $V(x+\eta)$ using a Taylor series:

$$V(x+\eta) = V(x) + \eta\frac{dV}{dx} + \frac{\eta^2}{2!}\frac{d^2V}{dx^2} + \cdots \qquad \text{\# expand Taylor series}$$

So now if we substitute these last two expressions into Equation 3 and group terms we see that

$$\mathcal{S}(x+\eta) = \int_{t_1}^{t_2} \left[\frac{1}{2}m\left(\frac{dx}{dt}\right)^2 - V(x) + \frac{1}{2}m\left(2\frac{dx}{dt}\frac{d\eta}{dt}\right) - \eta\frac{dV}{dx} - \text{higher order terms}\right] dt$$

[1] $V[x(t)]$ is sometimes called $U[x(t)]$ or simply $U(x)$, and is also informally referred to as PE.

We also have the constraint that

$$\delta\mathcal{S} = \mathcal{S}(x+\eta) - \mathcal{S}(x) = 0$$

and we saw earlier (Equation 2) that

$$\mathcal{S}(x) = \int_{t_1}^{t_2} \left[\frac{1}{2}m\left(\frac{dx}{dt}\right)^2 - V(x)\right] dt$$

Putting the last three equations together (and ignoring higher order terms in the Taylor series for $\mathcal{S}(x+\eta)$) we find that

$$\delta\mathcal{S} = \int_{t_1}^{t_2} \left[m\left(\frac{dx}{dt}\frac{d\eta}{dt}\right) - \eta\frac{dV}{dx}\right] dt = 0$$

What we would like is to have this integral be zero regardless of the value of η. So how can we "factor out" η? The first thing to notice is that by the product rule [6] we have

$$\begin{aligned}
\frac{d}{dx}\left(\frac{dx}{dt}\cdot\eta\right) &= \frac{d^2x}{dt^2}\cdot\eta + \frac{dx}{dt}\cdot\frac{d\eta}{dt} && \text{\# product rule: } \frac{d}{dx}(u\cdot v) = \frac{du}{dx}\cdot v + u\cdot\frac{dv}{dx} \\
\Rightarrow \frac{dx}{dt}\frac{d\eta}{dt} &= \frac{d}{dx}\left(\frac{dx}{dt}\eta\right) - \frac{d^2x}{dt^2}\eta && \text{\# solve for } \frac{dx}{dt}\frac{d\eta}{dt} \\
\Rightarrow \int_{t_1}^{t_2}\frac{dx}{dt}\frac{d\eta}{dt} &= \int_{t_1}^{t_2}\frac{d}{dx}\left(\frac{dx}{dt}\eta\right)dt - \int_{t_1}^{t_2}\frac{d^2x}{dt^2}\eta\, dt && \text{\# integrate both sides} \\
\Rightarrow \int_{t_1}^{t_2}\frac{dx}{dt}\frac{d\eta}{dt} &= \frac{dx}{dt}\eta\Big|_{t_1}^{t_2} - \int_{t_1}^{t_2}\frac{d^2x}{dt^2}\eta\, dt && \text{\# integral \& derivative in } 2^{nd} \text{ term cancel} \\
\Rightarrow \int_{t_1}^{t_2}\frac{dx}{dt}\frac{d\eta}{dt} &= -\int_{t_1}^{t_2}\frac{d^2x}{dt^2}\eta\, dt && \text{\# } \eta(t_1) = \eta(t_2) = 0 \text{ (by definition)}
\end{aligned}$$

So now we know that

$$\int_{t_1}^{t_2}\frac{dx}{dt}\frac{d\eta}{dt} = -\int_{t_1}^{t_2}\frac{d^2x}{dt^2}\eta\, dt \tag{4}$$

and we saw earlier that

$$\delta\mathcal{S} = \int_{t_1}^{t_2}\left[m\left(\frac{dx}{dt}\frac{d\eta}{dt}\right) - \frac{dV}{dx}\eta\right] dt = 0 \tag{5}$$

so

$$\begin{aligned}
\delta\mathcal{S} &= \int_{t_1}^{t_2}\left[-m\frac{d^2x}{dt^2}\eta - \frac{dV}{dx}\eta\right] dt && \text{\# substitute result from Equation 4 into Equation 5} \\
&= \int_{t_1}^{t_2}\left[-m\frac{d^2x}{dt^2} - \frac{dV}{dx}\right]\eta\, dt && \text{\# factor out } \eta
\end{aligned}$$

Since $\delta\mathcal{S}$ must equal zero (Equation 5) and we know $\eta(t)$ is not the constant function 0 it must be that

$$-m\frac{d^2x}{dt^2} - \frac{dV}{dx} = 0$$

But now note that

$$-\frac{dV}{dx} = m\frac{d^2x}{dt^2}$$

and since $-\dfrac{dV}{dx} = F$ and $\dfrac{d^2x}{dt^2} = a$ we amazingly have arrived at Newton's Second Law [4]:

$$F = ma$$

4 The Euler-Lagrange Equations

Consider the following seemingly counter-intuitive combination of the kinetic and potential energies (T and V respectively):

$$L = T - V \tag{6}$$

L here is called the *Lagrangian*. The Lagrangian is essentially the action per unit time. An application of the Lagrangian involves a mass on the end of a spring. In this case we have $T = \dfrac{m}{2}\dot{x}^2$ (where $\dot{x}$ is shorthand for $\frac{dx}{dt}$) and $V = \frac{k}{2}x^2$. So now we have

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2 \tag{7}$$

As we saw above (in slightly different notation) we can write

$$\frac{d}{dx}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x} \tag{8}$$

Equation 8 is called the *Euler-Lagrange* equation. For the mass on the end of a spring problem we looked at above we can see that from Equation 7 that $\frac{\partial L}{\partial \dot{x}} = m\dot{x}$ and $\frac{\partial L}{\partial x} = -kx$. Combining these results in Equation 8 gives us

$$m\ddot{x} = -kx$$

which is exactly the result obtained by using $F = ma$. An equation such as Equation 8 which is derived from the Euler-Lagrange equation, is called an equation of motion[2].

I'll just note here that in the general case the Lagrangian looks like this:

$$L = \frac{1}{2}m\dot{x}^2 - V(x)$$

The integral of L between two points t_1 and t_2 gives us the *action* of the path between those points (Equation 2).

Now, if we combine the Lagrangian (Equation 6) with the Euler-Lagrange equation (Equation 8) we get

$$m\ddot{x} = -\frac{dV}{dx}$$

But $-\frac{dV}{dx}$ is just the force on the particle. So as we saw at the end of Section 3 Equations 6 and 8 (albeit written in different notation) together again tell us that $F = ma$.

[2] AFAICT the term "equation of motion" is somewhat ambiguous. It is understood to refer to the second-order differential equation satisfied by x and not the actual equation for x as a function of t, namely $x(t) = A\cos(\omega t + \phi)$ in this problem, which is obtained by integrating the equation of motion twice.

5 Conclusions

It's amazing that two such different approaches lead to the same result...

References

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